

Suntan Cream

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P1

Pentyl 4-methoxycinnamate, **A**, is used in various suntan creams to absorb excessive ultra violet radiation and stop the skin burning.

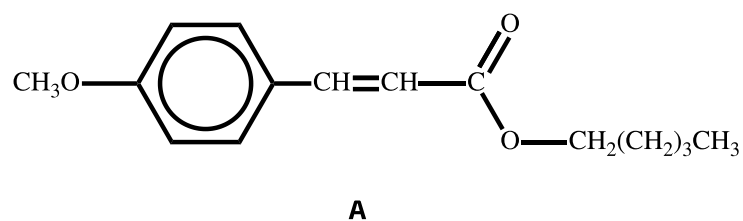


Figure 1: Structure of pentyl 4-methoxycinnamate, **A**

Part A

Isomers

The formula above represents two isomers. State the type of stereoisomerism **A** displays.

Part B

Structures of A

Use the [structure editor](#) to draw the two isomers and give their SMILES strings below in the format "**X**, **Y**" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

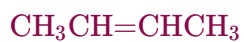
Question deck:

STEM SMART Chemistry Week 24

Isomerism in But-2-ene

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P1

Part A



Draw a diagram of each stereoisomer which exists with the structure $\text{CH}_3\text{CH}=\text{CHCH}_3$ in this external [structure editor](#).

When you have finished your structure click on the smiley face in the top left of the structure editor and copy and paste the string of letters (SMILES strings) into the box here.

Enter their structures as SMILES strings in the format "A, B" (space after comma).

[Using the structure editor](#)

Part B

Feature of the molecule

What feature of the molecule enables these two isomers to exist as separate entities?

Part C

Type of stereoisomerism

What is the name given to this type of stereoisomerism?

Adapted with permission from UCLES, Structured Science Scheme, June 1995, Unit C3: Essential Organic Chemistry, Question 4

Question deck:

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Cis-trans Isomerism

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P1

Part A

Cis-trans isomers

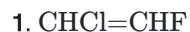
Which formula could represent a compound which has *cis-trans* isomers?

- ☐ $\text{C}_2\text{H}_2\text{Cl}_2$
- ☐ $\text{C}_2\text{H}_3\text{Cl}$
- ☐ $\text{C}_2\text{H}_2\text{O}_4$
- ☐ $\text{C}_2\text{H}_6\text{O}_2$

Part B

E-Z or cis-trans

Which of the following exhibit *E-Z* isomerism but **not** *cis-trans* isomerism?



- ☐ 1, 2 and 3 are correct
- ☐ 1 and 2 only are correct
- ☐ 2 and 3 only are correct
- ☐ 1 only is correct
- ☐ 3 only is correct

Part A adapted with permission from UCLES, A-Level Chemistry, November 1996, Paper 4, Question 19;

Part B created for isaacphysics.org by R. Less

Question deck:

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E-Z Isomerism

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P1

Part A

Pairs of isomers

Which of the following pairs illustrate *E-Z* isomerism?

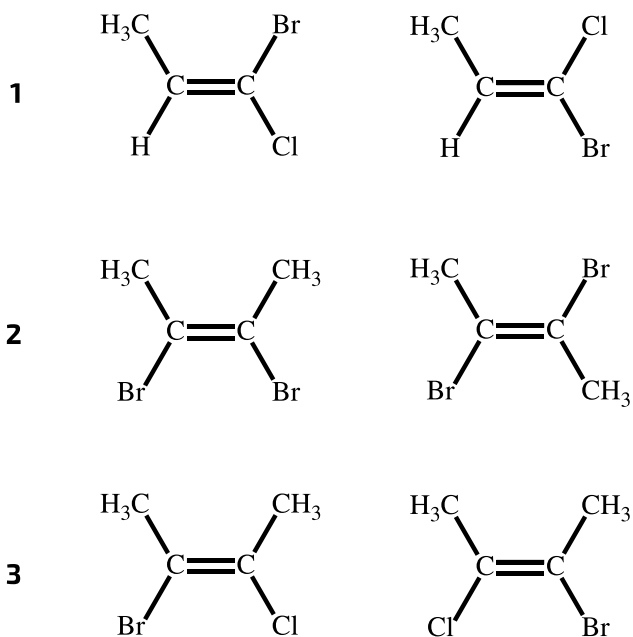


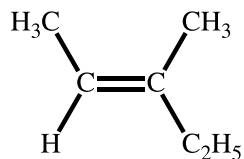
Figure 1: Pairs of stereoisomers

- ☐ 1, 2 and 3 are correct
- ☐ 1 and 2 only are correct
- ☐ 1 and 3 only are correct
- ☐ 2 and 3 only are correct
- ☐ 1 only is correct
- ☐ 2 only is correct
- ☐ 3 only is correct

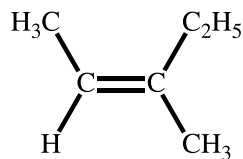
Part B

Isomers of C_6H_{12}

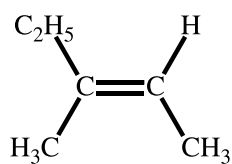
Four isomers of C_6H_{12} are shown below.



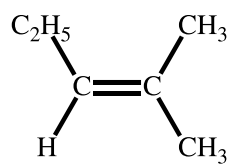
1



2



3



4

Figure 2: Four isomers of C_6H_{12}

Which of the following pairs consists of a pair of *cis-trans* isomers?

- ☐ 1 and 2
- ☐ 1 and 3
- ☐ 1 and 4
- ☐ 2 and 4
- ☐ 3 and 4

Part A adapted with permission from UCLES, A-Level Chemistry, November 1992, Paper 4, Question 38;

Part B adapted with permission from UCLES, A-Level Chemistry, June 1992, Paper 4, Question 21

Question deck:

STEM SMART Chemistry Week 24

Isomers of Hydrocarbons

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P1

Part A

Isomers of C_4H_{10}

Use the [structure editor](#) to draw all structural isomers of C_4H_{10} .

Give your answer as SMILES strings in the format "**A, B, etc.**" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part B

Isomers of C_5H_{12}

Use the [structure editor](#) to draw all structural isomers of C_5H_{12} .

Give your answer as SMILES strings in the format "**A, B, etc.**" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part C

Isomers of C_6H_{14}

How many structural isomers of C_6H_{14} are there?

Part D

Isomers of C_4H_8

How many structural isomers of C_4H_8 are there?

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Question deck:

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Isomers of Butanol

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P1

Alcohols can be classified as *primary*, *secondary* or *tertiary*.

Part A

Primary alcohols of formula $C_4H_{10}O$

Use the [structure editor](#) to draw all the isomers of the *primary* alcohols of formula $C_4H_{10}O$.

Give your answer as SMILES strings in the format "**A**, **B**, etc" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part B

Secondary alcohols of formula $C_4H_{10}O$

Use the [structure editor](#) to draw all the isomers of the *secondary* alcohols of formula $C_4H_{10}O$.

Give your answer as SMILES strings in the format "**A**, **B**, etc" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Part C

Tertiary alcohols of formula $C_4H_{10}O$

Use the [structure editor](#) to draw all the isomers of the *tertiary* alcohols of formula $C_4H_{10}O$.

Give your answer as SMILES strings in the format "**A**, **B**, etc" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Adapted with permission from UCLES, Modular Science, November 1996, Chains and Rings, Question 6

Question deck:

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Isomers of $C_5H_{10}O$

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level P2

The various structural isomers of $C_5H_{10}O$ can contain different functional groups.

Part A

Aldehydes

How many structural isomers of $C_5H_{10}O$ are **aldehydes**?

Part B

Ketones

Use the [structure editor](#) to draw all **ketones** of formula $C_5H_{10}O$.

Give your answer as SMILES strings in the format "**A, B, etc.**" (space after comma).

In the editor, after drawing your structure, click on the round, yellow smiley face to generate a SMILES string. Copy the SMILES string and paste it in the answer box.

[Using the structure editor](#)

Isomers of C_4H_8O

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level C1

Part A

Test with Tollens' reagent

How many structural isomers with the molecular formula C_4H_8O can reduce a solution containing $Ag(NH_3)_2^+$ ions (Tollens' reagent) to form a silver mirror?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Part B

Containing C=O group

How many structural isomers with the molecular formula C_4H_8O contain the C=O group?

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5

Part A adapted with permission from UCLES, A-Level Chemistry, November 1996, Paper 4, Question 275;

Part B created for isaacphysics.org by R. Less

Question deck:

STEM SMART Chemistry Week 24

Oxygen-inserting Bacteria

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level C1

Bacteria have been suggested as a possible means of cleaning up oil spillages. Some bacteria contain enzymes that can insert one or more oxygen atoms into any carbon-hydrogen bond in an alkane. This converts a water-insoluble alkane into a water-soluble alcohol, e.g. $\text{CH}_3\text{CH}_3 \longrightarrow \text{CH}_3\text{CH}_2\text{OH}$.

Which of the following alcohols could be obtained by this process from $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3$?

1 $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$

2 $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{CH}_3)_2$

3 $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_2\text{OH})_2$

- ☐ **1, 2 and 3** are correct
- ☐ **1 and 2** only are correct
- ☐ **1 and 3** only are correct
- ☐ **2 and 3** only are correct
- ☐ **1** only is correct
- ☐ **2** only is correct
- ☐ **3** only is correct

Adapted with permission from UCLES, A-Level Chemistry, November 1995, Paper 4, Question 39

Question deck:

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Isomers of $C_2H_4O_2$

Subject & topics: Chemistry | Organic | Isomerism **Stage & difficulty:** A Level C1

The isomers of $C_2H_4O_2$ can contain different functional groups. Draw the structures of **all** the isomers of $C_2H_4O_2$ which contain the following functional groups and do not contain an O—O bond.

Use the [structure editor](#) to generate SMILES strings.

Enter their structures as SMILES strings in the format "**A**, **B**, ... etc." (space after commas).

Part A

Carbonyl group

Part B

Alkene group

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Question deck:

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